University of Michigan

NERS 515

Nuclear Measurements Laboratory

Laboratory 2: Single-Channel and Multichannel Pulse Analysis

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Abstract

This experiment evaluates the fundamental operating characteristics of a Single-Channel Analyzer (SCA) and a Multichannel Analyzer (MCA) using simulated detector pulses. By varying the amplitude of a pulse generator, the integral linearity and discrimination boundaries of the SCA were mapped across its integrator, window, and normal operating modes. The SCA discriminator limits tracked the input amplitudes measured by the oscilloscope linearly across the tested range. Furthermore, the MCA demonstrated a proportional relationship between the input pulse height and the output channel, which produced a calibration of approximately 766 channels/V. Finally, a double-pulse methodwas used to measure a twin-pulse dead time of 3.68 μs at an estimated channel number of 3780.

1 Introduction

In radiation spectroscopy, the electrical charge generated by an interaction must be processed and sorted to determine the energy of the incident particle [1]. This experiment introduces us to the primary electronic modules responsible for this signal processing: the Single-Channel Analyzer (SCA), which filters pulses based on specific amplitude thresholds, and the Multichannel Analyzer (MCA), which digitizes and sorts pulses into a differential amplitude spectrum. Characterizing the input-output linearity, shaping behavior, and dead-time limitations of these modules is essential, as these parameters directly impact energy calibration and counting accuracy in high-rate environments.

2 Theory

When radiation interacts within a detector, it produces a burst of charge Q. This charge is momentarily integrated across the equivalent circuit capacitance C to produce a voltage pulse with a maximum amplitude given by $V_{max} = Q/C$ (Knoll, Eq. 4.9) [1].

Because these preamplifier tail pulses have long decay times, a linear shaping amplifier is required to reduce pulse overlap (pile-up) and optimize the signal-to-noise ratio [1]. Shaping is commonly performed using a CR-RC network. Knoll defines the output waveform E_{out} for a step input E passed

through a CR-RC network with equal differentiation and integration time constants ($\tau = RC$) as $E_{out} = E\frac{t}{\tau}e^{-t/\tau}$ (Knoll Eq. 17.22) [1].

Once shaped, the pulses are analyzed by discriminators. An integral discriminator generates a logic pulse only if the analog signal exceeds a fixed threshold [1]. An SCA operates as a differential discriminator by establishing an acceptance window between a lower-level and an upper-level discriminator; it outputs a logic pulse only if the input amplitude falls within this window [1]. The MCA expands on this by using an Analog-to-Digital Converter (ADC) to sort pulses into discrete channels. The theoretical calibration for mapping a 0-10 V input range across an 8192-channel MCA is calculated as $8192/10 \approx 819$ channels/V.

At high counting rates, the system's dead time τ (the minimum time required to separate two events) causes counting losses. Knoll models dead time behavior in two extremes. For a nonparalyzable system, the true event rate n relates to the measured rate m by $n = \frac{m}{1-m\tau}$ (Knoll, Eq. 4.24) [1]. For a paralyzable system, where events during the dead period continuously extend the recovery time, the relationship is $m = ne^{-n\tau}$ (Knoll, Eq. 4.27) [1].

3 Equipment and Procedure

3.1 Equipment

Details of all equipment used in this lab are listed in Table 1. The list does not include the SCA because we were advised not to remove the instrument from the NIM in case it caused other issues.

Purpose	**Manufacturer**	**Model**	**Serial Number**
Oscilloscope	Agilent Technologies	DSO6052A	MY44007040
Tail Pulse Generator	BNC	BH-1	32412
Amplifier	Ortec	572A	121
Multi-Channel Analyzer	Ortec	Easy-MCA	15170150

Table 1: List of equipment used in Laboratory 2.

3.2 Experimental Procedure

We began the procedure by connecting the output of a pulse generator to the input of a linear amplifier, which was set to a fixed gain of 20. We split the amplifier output, routing one branch to one channel of the oscilloscope and the other branch to the input of the SCA. The logic output of the SCA was then connected to a second channel of the oscilloscope to monitor the signals. With the SCA set to integrator mode, it functioned as a simple lower-level discriminator (LLD). This setup is visualized in Figure **??**. For various pulser amplitudes, we adjusted the LLD dial to determine the maximum threshold setting that still generated an output logic pulse. Using the oscilloscope's cursor function, we measured the input pulse amplitude as the difference between the baseline and the peak. We then measured the pulse rise time, utilizing the cursors as the time required for the pulse to reach its peak amplitude from 37% of that maximum [2].

Later on, we evaluated the SCA in window mode. After selecting three representative LLD settings, we varied the window width dial for each and recorded the minimum and maximum pulser amplitudes that successfully triggered an SCA output pulse. We then switched the SCA to normal mode, which utilizes an independent upper-level discriminator (ULD) to set the window. Choosing three different representative LLD settings, we varied the ULD dial and again recorded the bounding input pulse amplitudes that resulted in a logic pulse [2].

To characterize the MCA, we routed the shaped amplifier output directly to the MCA input, ensuring the signals met the 0–10 V positive input requirement of the device. This setup is visualized in Figure **??**. The pulser frequency was set to approximately 1 kHz. We tested three different amplifier gains, applying three distinct pulser amplitudes for each gain setting. We acquired data for a fixed live time without clearing the MCA memory between steps. This procedure produced a single cumulative spectrum containing multiple isolated peaks. With all device variables set, we measured the net counts produced by the pulser [2].

Lastly, we measured the MCA dead time by utilizing the pulse generator's double-pulse feature. We supplied the shaped twin pulses to both the oscilloscope and the MCA. Starting with a minimum delay, we slowly increased the temporal spacing between the twin pulses until the MCA resolved a distinct second peak, at which point we measured this critical time separation on the oscilloscope [2]. Due to time constraints, we deviated from

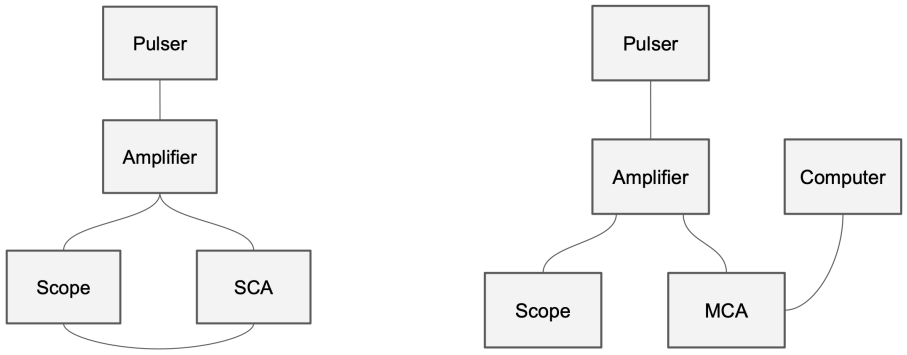


Figure 1: Equipment setup block diagrams for (a) SCA and (b) MCA measurements.

the manual's requested procedure and measured one data point rather than the multiple amplitudes required to construct a complete dead-time versus channel-number plot.

4 Data

4.1 Single-Channel Analyzer: Integrator LLD Mode

The results following Experiment 2.1 are summarized in Table 2. The amplitude V_m was measured in the oscilloscope after the pulse amplitude V_0 is amplified by the amplifier gain = 20. The amplitude V_m was calculated by subtracting the base and peak amplitudes on the scope, performed automatically by the scope using the cursor function. We did not record these raw baseline and peak values, only its difference, so we cannot provide them in this report. However, we did calculate the error σ_m using the error propagation formulas in Knoll Chapter 3, which is shown in Section 5.1.

We calculate the rise time as the time the pulse takes to reach V_m from 37% of V_m. We obtained the error for each time measurement by measuring half the time width where $0.37V_m$ and V_m ranges within its error ($\sigma_m = 25.4$ mV). The error calculation is shown in Section 5.1.

Pulser Amplitude V_0	Scope Amplitude V_m	Rise Time	LLD_{min}	LLD_{max}
100 mV	3150 ± 25.4 mV	684 ± 86.0 ns	0.0 V	3.00 V
1.1 V	5125 ± 25.4 mV	710 ± 86.0 ns	0.0 V	5.62 V
1.8 V	7175 ± 25.4 mV	660 ± 86.0 ns	0.0 V	7.05 V
2.1 V	7875 ± 25.4 mV	780 ± 86.0 ns	0.0 V	7.70 V
3.1 V	9975 ± 25.4 mV	680 ± 86.0 ns	0.0 V	9.90 V
0.26 V	62 ± 25.4 mV	660 ± 86.0 ns	0.0 V	0.025 V

Table 2: Minimum and maximum LLD limits for various pulser amplitudes with amplifier gain = 20, SCA in *integrator* mode.

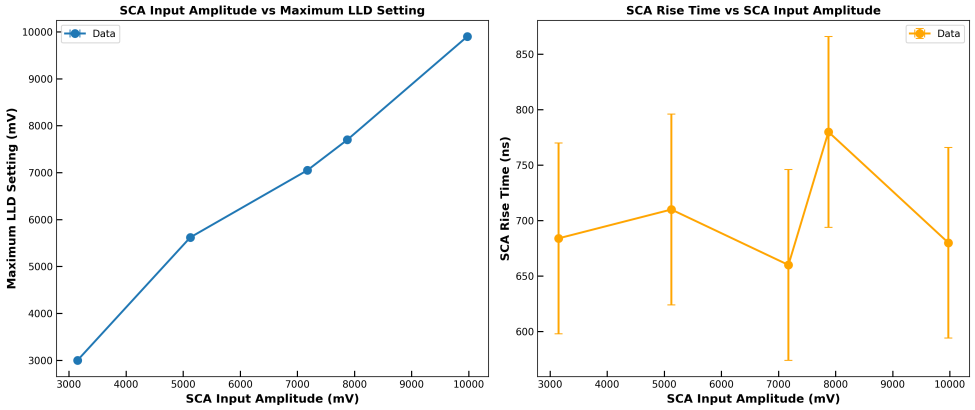


Figure 2: (a) Maximum LLD setting vs. scope-measured input amplitude V_m. (b) SCA rise time vs. V_m.

4.2 Single-Channel Analyzer: Window and Normal Modes

The results from experiment 2.2 SCA Window and Normal Modes are given in Table 3 and 4 and plotted in Figures 3 and 4, respectively. We chose three representative LLD settings for each mode: 4.0 V, 6.83 V, and 8.0 V for window mode, and 2.5 V, 4.0 V, and 6.5 V for normal mode.

LLD	Window Dial Setting	V_{min}	V_{max}
6.83 V	2.4	5.83 V	5.99 V
	3.1	5.83 V	6.05 V
	4.5	5.83 V	6.17 V
8.00 V	2.4	6.84 V	6.98 V
	3.1	6.84 V	7.05 V
	4.5	6.84 V	7.16 V
4.00 V	2.4	3.40 V	3.56 V
	3.1	3.40 V	3.62 V
	4.5	3.40 V	3.75 V

Table 3: Minimum and maximum discriminated pulse amplitudes as a function of window size, with amplifier gain = 4, SCA in *window* mode, for three LLD settings.

LLD	Upper Limit	V_{min}	V_{max}
4.0 V	5.0 V	3.40 V	4.18 V
	6.0 V	3.40 V	5.05 V
	8.5 V	3.40 V	7.22 V
2.5 V	4.0 V	2.09 V	3.32 V
	6.0 V	2.09 V	5.12 V
	10.0 V	2.09 V	8.53 V
6.5 V	7.0 V	5.55 V	5.93 V
	8.6 V	5.55 V	7.29 V
	10.0 V	5.55 V	8.53 V

Table 4: Minimum and maximum discriminated pulse amplitudes as a function of the upper-level setting, with amplifier gain = 4, SCA in *normal* mode, for three LLD settings.

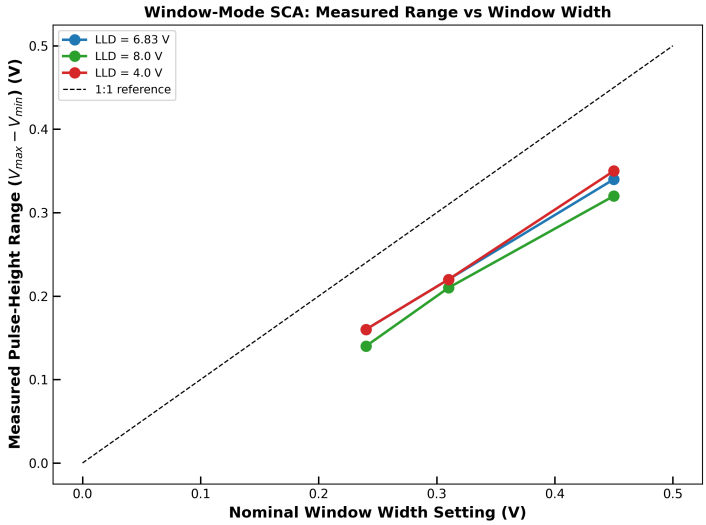


Figure 3: Measured pulse-height range ($V_{max} - V_{min}$) vs. nominal window-width setting (dial reading $\div$ 10), for three LLD values, with a 1:1 reference line.

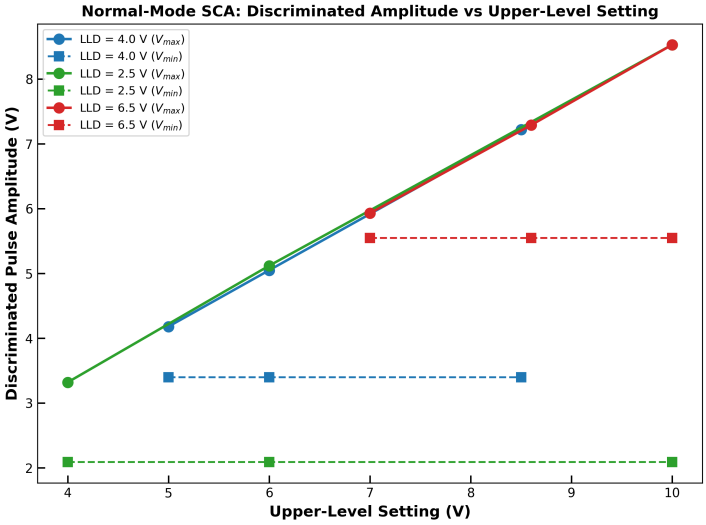


Figure 4: Discriminated pulse amplitude (V_{min}, dashed; V_{max}, solid) as a function of the upper-level setting, grouped by LLD.

4.3 Multichannel Analyzer: General Measurements

For three amplifier gains (10, 11.8, and 13.6), we recorded three pulser amplitudes each without clearing the MCA memory between steps, producing a spectrum with one isolated peak per pulser amplitude with nine peaks total. Table 5 shows the pulser amplitude V_0, the scope-measured amplifier output amplitude, the data acquisition time, and the MCA channel number of the resulting peak. We assume the error on each measured amplitude to be $\sigma_m = 25.4$ mV, the same value obtained in Section 5.1 for the integrator-mode measurements, since the same oscilloscope and cursor method were used. All runs were acquired with a 30 s live time except for the first (gain = 10, $V_0 = 1.0$ V), which was acquired for 60 s.

Gain	Pulser Amplitude V_0	Measured Amplitude	Live Time	Peak Channel #
10	1.0 V	2.62 ± 0.025 V	60 s	2009
	3.0 V	4.68 ± 0.025 V	30 s	3619
	0.5 V	671 ± 25.4 mV	30 s	558
11.8	820 mV	1.3 ± 0.025 V	30 s	1021
	1.5 V	2.3 ± 0.025 V	30 s	1787
	2.71 V	3.9 ± 0.025 V	30 s	3022
13.6	28.6 mV	496.2 ± 25.4 mV	30 s	406
	4.1 V	7.04 ± 0.025 V	30 s	1236
	1.5 V	2.7 ± 0.025 V	30 s	2093

Table 5: MCA peak channel number for nine pulser-amplitude/amplifier-gain combinations.

Figure 5 shows the raw recorded spectra (counts vs. channel number, zoomed to the peak regions) for all nine runs, grouped by amplifier gain, reproducing what was displayed in the ORTEC Maestro software during acquisition since we did not record a direct screenshot.

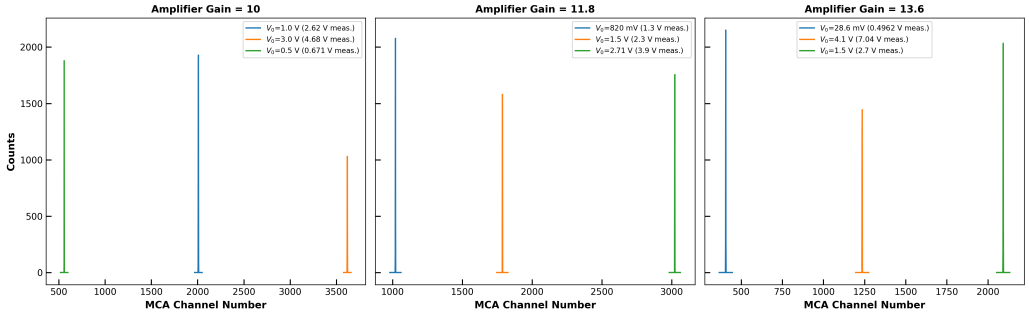


Figure 5: Recreated MCA spectra (counts vs. channel number, zoomed to the peak region) for the nine runs of Table 5, grouped by amplifier gain.

4.4 Multichannel Analyzer: Twin-Pulse Dead-Time Measurement

We report a single measurement in Table 6, but note explicitly that it is not sufficient on its own to produce the dead-time-vs.-channel-number plot requested by the manual (addressed further in Section 5.4).

Gain	Measured Amplitude	Estimated Channel #	MCA Dead Time
13.6	4.9 ± 0.025 V	$\sim$3780	3.68 μs

Table 6: Twin-pulse dead-time measurement. Only a single initial amplitude was recorded, so no dead-time-vs.-channel-number trend can be constructed from this data alone.

5 Analysis of Data

5.1 SCA Integrator LLD Mode

We took the error as half of the measured width of the lines at the base ($\sigma_b = 25$ mV) and peak ($\sigma_p = 4.6875$ mV) lines [1]. We assume σ_b and σ_p are the same across all measurements for this experiment since we use the

same equipment and cursor method for each. The amplitude error is then:

$$\sigma_m = \sqrt{\sigma_b^2 + \sigma_p^2}$$
$$\sigma_m = \sqrt{25^2 + 4.6875^2}$$
$$\sigma_m = 25.4 \text{ mV}$$

For the same reason, we assume the time-cursor error is the same across all measurements. For the $V_0 = 1.1$ V trial, the pulse reached $0.37V_m$ at $t_1 = -1700$ ns, with error $\sigma_{t1} = 2.7$ ns, and reached V_m at $t_2 = -990$ ns, with error $\sigma_{t2} = 86$ ns. The rise time and its error are then:

$$t_{rise} = t_2 - t_1 = -990 - (-1700) = 710 \text{ ns}$$
$$\sigma_{rise} = \sqrt{\sigma_{t1}^2 + \sigma_{t2}^2}$$
$$\sigma_{rise} = \sqrt{2.7^2 + 86^2}$$
$$\sigma_{rise} = 86.0 \text{ ns}$$

This same rise-time error of 86.0 ns was applied to all rows of Table 2.

In integrator mode, the SCA is expected to behave as a simple lower-level discriminator. It outputs a logic pulse for any input pulse whose amplitude exceeds the LLD setting, so the maximum LLD at which the SCA still fires should be the peak amplitude of the input pulse. Figure 2a shows that the maximum LLD setting increases almost linearly with the scope-measured input amplitude V_m (Table 2), with LLD_{max} tracking V_m to within the measurement uncertainty across the full range tested ($\approx 0 \rightarrow 10$V). This is consistent with the expected behavior and indicates that the SCA threshold is well-calibrated against the measured amplitude over more than two orders of magnitude in pulse height. Figure 2b shows that the SCA rise time remains roughly constant (660–780 ns) regardless of input amplitude, as expected, since rise time is set by the shaping electronics of the SCA input stage rather than the amplitude of the pulse being discriminated, and the error bars in both panels are consistent with this interpretation. We note that LLD_{min} is 0.0 V for every row of Table 2, reflecting that any pulse above the baseline fires the SCA once the LLD is returned to its lowest setting.

5.2 SCA Window and Normal Modes

Figure 3 shows the measured pulse-height range ($V_{max} - V_{min}$) plotted against the nominal window-width setting for all three LLD values with a 1:1 line.

The measured range increases linearly with the nominal window-width setting for all three LLD groups, confirming that the window dial behaves linearly over the range tested. However, the measured range consistently falls below the 1:1 line. For example, the 0.45 V window setting yields a measured range of only 0.32–0.35 V. This systematic offset suggests either that the stated dial-to-volts conversion factor is only approximate for this particular SCA unit, or that there is some other additional constant offset in how the window boundary is defined. However, it does not affect the underlying linearity of the window control.

Figure 4 shows that in normal mode, V_{max} increases with the upper-level setting for all three LLD groups, while V_{min} remains near the corresponding LLD value, consistent with the SCA passing only pulses between the LLD and ULD settings. We note that the measured V_{max} values are consistently slightly below the nominal upper-level setting. As in window mode, this is a small, systematic offset between the dial reading and the true discriminated edge and does not affect the qualitative linear trend. Together, Figures 3 and 4 confirm the SCA's operating principle as an adjustable bandpass discriminator, consistent with the single-threshold behavior observed in integrator mode (Figure 2).

5.3 MCA Linearity

Table 7 reports the net counts recorded under each peak of Table 5, extracted directly from the corresponding `.Chn` spectrum files. All runs were acquired with a 30 s live time except for the first (gain = 10, $V_0 = 1.0$ V), which was acquired for 60 s, so its raw count is not directly comparable to the other eight without normalizing by live time. Dividing each run's net counts by its live time gives a count rate of 89.1 counts/s for every run, including the 60 s run, confirming that the pulser's trigger rate was unaffected by the changes in amplifier gain and pulser amplitude between runs.

Gain	Pulser Amplitude V_0	Measured Amplitude	Net Counts
10	1.0 V	2.62 ± 0.025 V	5346
	3.0 V	4.68 ± 0.025 V	2673
	0.5 V	671 ± 25.4 mV	2673
11.8	820 mV	1.3 ± 0.025 V	2673
	1.5 V	2.3 ± 0.025 V	2673
	2.71 V	3.9 ± 0.025 V	2673
13.6	28.6 mV	496.2 ± 25.4 mV	2673
	4.1 V	7.04 ± 0.025 V	2673
	1.5 V	2.7 ± 0.025 V	2673

Table 7: Net counts recorded under each MCA peak of Table 5.

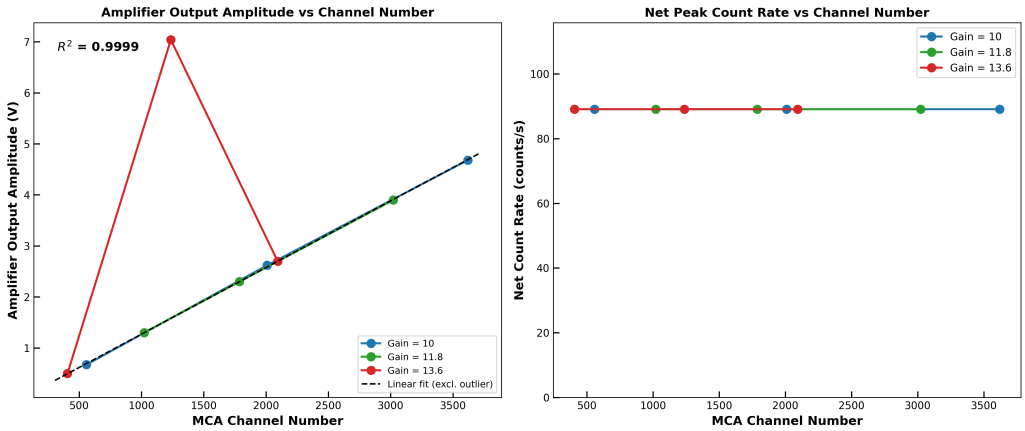


Figure 6: (a) Amplifier output amplitude vs. MCA channel number, with a linear fit. (b) Net count rate vs. MCA channel number. Both panels show the three amplifier gains of Table 5.

Figure 6(a) shows that, for seven of the nine runs in Table 5, the MCA channel number of the recorded peak increases linearly with the amplifier output amplitude, as expected for an ADC whose output channel is proportional to input pulse height. A linear fit to these points (excluding the gain = 13.6, $V_0 = 4.1$ V run discussed below) gives channel $= 765.8 \times V + 27.6$, with $R^2 = 0.9999$, confirming excellent linearity over the tested range. In other words, for every 1 V increase in pulse amplitude, the peak moves about

766 channels higher, with an initial offset of 28. With 8192 total channels, the 28 channel intercept is small and close to zero. This is consistent with the MCA's full 0–10 V input range being mapped onto its 8192-channel span ($8192/10 \approx 819$ channels/V). The run with gain = 13.6, $V_0 = 4.1$ V (recorded at channel 1236) is a clear outlier. Based on the fitted calibration, a 7.04 V pulse should appear near channel 5420, not channel 1236. This suggests either a transcription error in the recorded "measured amplitude" for this run, or a change in an attenuation/conversion-gain setting between this run and the others.

Figure 6(b) shows that, once normalized by live time, all nine runs give the same count rate (89.1 counts/s), as reported in Table 7. This confirms that the pulser's trigger rate was unaffected by the changes in amplifier gain and pulser amplitude between runs, and that our normalization correctly accounts for the first run's longer live time.

5.4 MCA Dead Time

Following the double-pulse procedure described in the manual, at amplifier gain 13.6 an initial pulse with measured (amplified) amplitude 4.9 V required a pulse spacing of 3.68 μs before the MCA resolved the second, delayed pulse as a distinct peak. Applying the channel calibration obtained in Section 5.3, this amplitude corresponds to an initial-pulse channel number of approximately 3780.

For the twin-pulse dead-time measurement (Section 4.4), only a single initial-amplitude/dead-time pair was recorded (3.68 μs at an estimated channel number of $\sim$3780), so we cannot construct the dead-time-vs.-channel-number trend requested by the manual, which is needed to answer Post-Lab Question 4.

5.5 Post-Lab Questions

1. Why is integral linearity important? Good integral linearity allows one calibration to be trusted across the entire amplitude range of interest. Poor integral linearity would mean that a calibration performed at one amplitude could not be reliably assumed for another, undermining spectroscopy applications.

2. How many SCAs would you need to equal the discriminatory capability of your specific MCA? Since an MCA with N channels sorts

pulses into N independent amplitude bins and our model has 8192 channels, you would need 8192 SCA channels to replicate its discriminatory capability.

3. What are the advantages and disadvantages of increasing the number of channels? More channels improve amplitude resolution, however each channel receives fewer counts on average. This increases the uncertainty per channel, requiring longer acquisition times to achieve the same statistical precision.

4. Based on your results from the plot in Section 3.2.2, what type of ADC is in your MCA? As discussed above, we currently have only one dead-time measurement and cannot yet distinguish between ADC types.

5. Explain ballistic deficit and pulse pileup. Can you minimize the problems associated with both of these effects at the same time? Why or why not? Ballistic deficit occurs when the amplifier shaping time is too short relative to the charge-collection time, causing a distorted shaped pulse. Pulse pileup occurs when two pulses arrive close enough together that their shaped tails overlap. Ballistic deficit is reduced by using a *longer* shaping time, so each pulse returns to baseline before the next one arrives. Increasing the shaping time to minimize ballistic deficit worsens pulse pileup and vice versa, so both effects cannot be minimized with a single fixed shaping time.

6 Conclusions

This laboratory successfully demonstrated most of the operations of the laboratory's SCA and MCA while familiarizing us with its use. The single-channel analyzer behaved as expected in all three operating modes: the integrator-mode LLD tracked the input pulse amplitude linearly (Figure 2), and both the window- and normal-mode discriminated ranges tracked their respective window-size and upper-level settings linearly while holding V_{min} fixed at the LLD (Figures 3 and 4), confirming the SCA's function as an adjustable bandpass pulse-height discriminator. The multichannel analyzer showed the expected linear relationship between input pulse amplitude and output channel number for most of the runs recorded, aside from the one outlier. With only one twin-pulse dead-time measurement, the type of ADC in our MCA cannot be determined.

Future work for this laboratory would be to collect additional dead-time

measurements to determine the type of ADC in our MCA. In addition, in future laboratories, we will record more raw measurements to supplement the calculated values for a more structured and comprehensive report, rather than relying solely on scope tools (e.g. including baseline and peak values for the amplitude calculation).

References

[1] G. F. Knoll, *Radiation Detection and Measurement*. Hoboken, NJ: John Wiley & Sons, Inc., fourth ed., 2010.

[2] University of Michigan, Department of Nuclear Engineering and Radiological Sciences, *NERS 515 Nuclear Measurements Laboratory: Laboratory 2*, 2026.